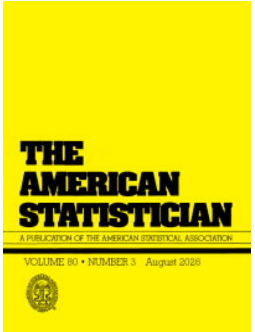

The American Statistician

ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: www.tandfonline.com/journals/utas20

Estimating Mean and Variance from Right-Censored Data

Dragan Banjevic

THE AMERICAN STATISTICIAN
2026, VOL. 00, NO. 0, 1–14: General
https://doi.org/10.1080/00031305.2026.2701333

Taylor & Francis
by informa

Check for updates

Estimating Mean and Variance from Right-Censored Data

Dragan Banjevic

Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Canada

ABSTRACT

Estimation of the population mean and variance from (right) censored data has gained more interest recently in biomedical and engineering studies. The purpose of the paper is to introduce some new ideas, but also to give a fresh look at the past that can be of interest in teaching, using an elementary approach. A simple intuitive procedure of imputation of censored values for estimation of the mean and variance, inspired by mean remaining life, has been established, and can be easily performed using a convenient recursive calculation. This method is strongly connected with the definition of the Kaplan-Meier survival curve. Rationale behind the K-M "mysterious" estimator of the variance of the sample mean has been discussed in elementary terms. The recursive procedure has been further extended to these cases. An approach to a measure of the effect of censoring has been suggested.

ARTICLE HISTORY

Received May 2025
Accepted July 2026

KEYWORDS

Population mean; Population variance; Recursive calculation; Remaining life; Right-censored data; Survival curve

1. Introduction

Every student, even in a high school, knows how to estimate (calculate) mean and variance from a complete sample. Unfortunately, students in elementary undergraduate courses are not even aware that a sample can be "incomplete." But when you step into a more realistic life problems almost all samples are "incomplete" when they come from observing lifetimes/failure times in biomedical and engineering studies, at least. Incompleteness of a sample usually comes from censoring/suspension due to observational time window or removal due to unrelated causes (e.g., a death due to a traffic accident, or planned time removals in engineering). It also can come from intermittent observations ("interval" censoring) and many other causes. The most common and simplest incompleteness is of the "right" censoring, where the observed lifetimes are "cut" from the right, that is, not observed till the end. One of the reasons for not considering censored observations sooner in regular teaching may be in complexity of a stochastic model required to describe data with censoring, often not easy to justify in practice, and a complex mathematics behind it. The main object of interest in lifetime analysis, the "survival function" (or "survivor function"/"survival curve") in biomedical studies, or the "reliability function" in engineering, is nonparametrically most often estimated by the Kaplan-Meier estimator. Alternatively, the Nelson-Aalen estimator is used (in the counting process approach), or the life tables estimator (in actuarial analysis), but we will not consider them here. Estimation of the population mean life μ is mentioned in passing in biomedical studies, although is not neglected in research, but little about the population variance appears, as it is either of no interest, or is considered a trivial problem. But, in engineering, both the mean and variance are of interest. It is also of interest to estimate the variance of the sample mean, as it is needed to calculate the estimation error; this problem is usually considered along with the mean life estimation. The variance of the "K-M mean" is estimated by another K-M type estimator. These estimators and their basic properties are introduced in the key historical Kaplan and Meier (1958) paper (the K-M paper, for short), which is still widely cited (the paper had 26,000 citations before July 2005 (Ryan and Woodall 2005), 62,200 in November 2022, and 68,700 at the end of December 2025, and counting!). The paper is written in somewhat "old" style, but is still interesting, instructive, and useful. We will consider some of its basic ideas here. It is not widely known that the K-M paper originated from a Meier's AMS conference paper in 1954. The main purpose of our paper is to introduce estimation of the mean and variance from right-censored data in a simple way, which can essentially be reduced to estimating the "remaining useful life" (RUL), in engineering, or the "mean residual life" (MRL), in biomedical studies. We must admit that interest in MRL becomes much larger nowadays, for example, as a tool to estimate prolonged life after a medical procedure. The method used in this paper applies very simple recursive calculation convenient in all estimations from right-censored data. The paper is intended to be self contained, it requires only a minimal knowledge of basic notions, and is not overly technical. All proofs are elementary, and do not require any stochastic model, but may not be trivial. An exception is Section 7, where a simple stochastic model is introduced. To avoid overburdening the reader, we placed all technical proofs in supplementary material. Using MRL is not complete novelty. For example, Datta (2005) describes three different methods of estimating the mean life (the K-M type integral from survival

CONTACT Dragan Banjevic banjev@mie.utoronto.ca Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Canada.
Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/TAS.

function, imputation using MRL, and "reweighting") and shows their equivalence without any insights in their rationale. All three methods are covered in our paper as a consequence of our initial simple intuitive idea of recursively imputing a censored value with the average of the values above it (that also may be imputed), without formally introducing MRL. Technically speaking, our final effective procedure with the "t^*" sequence (introduced in Section 3) does not use imputation, it calculates MRL recursively, without a need to define it formally, with the sample mean as the final result. Unlike Datta's approach, our method does not require the survival curve as an input; instead, it can be used to derive it. The "t^*" sequence is instructive and useful in itself, as it also gives "the total expected life" (see Note 4). Similar method is used for estimation of the variance. To justify and expand estimation of the variance of the K-M estimator of the mean we have introduced a relatively simple stochastic model in Section 7, following in essence the K-M paper. Using much more advanced theory of counting processes explained in brief in Klein and Moeschberger (2003), or in detail in Aalen, Borgan, and Gjessing (2008) and in Fleming and Harrington (2005) would help but would be hard for "lay" people. It should be clear that our stochastic model is simple and has shortcomings (see discussion in Section 7), but is still beneficial for introduction and basic understanding, and can give an initial background for some innovations. As mentioned above, work on the K–M type of calculation has proliferated, and a comprehensive review of even the most recent literature would be prohibitively long. Here we provide only a very brief overview, focusing on the topics most relevant to our paper and avoiding more technical ones. Only some works that use purely nonparametric methods are included (e.g., those using regression or semiparametric methods are not). Imputation in censored data is considered, for example, by Taylor, Murray, and Hsu (2002), Cole et al. (2020), De Felice et al. (2022), De Felice, Mazzoni, and Moriconi (2023), and Bouaziz (2023). Many papers are devoted to usage, adjustment and improvement of the K-M survival curve and its variance, among them Satten and Datta (2001), Borkowf (2005), Gillespie et al. (2010), Malla and Mukerjee (2010), Mukangai and Odongo (2016), Gu, Fan, and Yin (2022), Khan et al. (2022), Liang (2022), Qin, Sasinowska, and Leemis (2023), and Wu and Kolassa (2024). They often include some heuristic or ad-hoc approaches. Papers on "restricted mean life" and MRL are on rise, for example, Datta (2005), Qin and Zhao (2007), Shen, Xie, and Tang (2010), Zhao, Cheng, and Bang (2011) Royston and Parmar (2013), Alvarez-Iglesias et al. (2015), Paukner and Chappell (2021), and Wu, Kolassa, and Dong (2025). A lot of other papers use semi-parametric or parametric methods. The population variance is barely mentioned, except in some applications.

In Section 2, we introduce the idea of "MRL imputation" for censored observations and show its equivalence with the above mentioned Datta's "reweighting" method (originated by Efron 1967). In Section 3, we introduce a recursive type formula for the calculation of the mean estimator. In Section 4, we introduce the well-known K-M type estimator (the area under K-M survival function) for μ and show its equivalence with the simple recursive MRL type method using the "t^*" sequence. In Section 5, we introduce two types of variance estimation, one coming directly from the t^* sequence and the other of the K-M integral type and discuss their relationship. In Section 6, we introduce the well-known K-M estimator of the variance of the sample mean and show its relationship with the t^* sequence. Dependence of the K-M estimator on censoring is discussed. An example from the literature is used for illustration throughout the paper. In a somewhat more theoretical Section 7, a stochastic model for the data with censoring is introduced to allow for deriving the theoretical formula for the variance of the sample mean and to justify definition of its K-M estimator. This rationale is quite obscured in commonly used books and papers. That section is inspired by the original K-M paper and our recursive method, which greatly simplifies the derivation. In Section 8, a small simulation study (using R) is presented to illustrate the concepts discussed in the paper, including Section 7. Conclusions are given in Section 9. To keep the main presentation easy to follow, additional details and comments are provided in separate notes embedded in the text. Three Appendices are given in supplementary material, including technical proofs (referred as "Proof 1, 2, …") and a necessary short primer in Riemann-Stieltjes integration, with an addition of the R code used for simulation in Section 8. We believe that the material in this paper can help as a teaching tool for introductory fourth-year courses on survival analysis in biomedical or actuarial studies (Sections 1–6), as well as for graduate courses that incorporate research components (particularly Sections 7 and 8). Furthermore, all recursive calculations can be easily performed in spreadsheet form in Excel. Extending the provided R code to other censoring models would also be beneficial.

2. Censored Data Problem and Population Mean Estimation by Imputation

Let $0 \leq t'_1 \leq t'_2 \leq \cdots \leq t'_n$ be observed values of the lifetime variable T, either completed (failures) or right censored. Let their censoring indicators be $\delta_1, \delta_2, ., \delta_n$, where $\delta_i = 1$ if t'_i is a failure lifetime, and $\delta_i = 0$ if t'_i is a censored lifetime (strictly speaking, δ_i is the *failure* indicator, and $\delta'_i = 1 - \delta_i$ is the *censoring* indicator). When a censored lifetime is equal to a failed lifetime, it is placed in order *after* the failed lifetime, so the equal censored and failed lifetimes are not mixed.

To estimate the mean lifetime $\mu = \mu_T$ (or meant time to failure—MTTF), we can use the sample mean $\overline{t'} = \frac{1}{n}\sum_1^n t'_i$ if all lifetimes are failures. If some of times are censored, the first, naïve approach would be to use *only* the failed lifetimes (we have seen it in some engineering studies), but with it we would have a biased estimator (usually underestimation), as the lifetimes are cut from the right (very often the longest ones are censored). What can we do? Instead of fixing the estimator directly, we can try to "fix" the data itself. For censored times t'_i we can use some reasonable adjustment t_i^+, by imputing a higher value, and then take the simple average of such adjusted values. Let y_i be such "fixed" value, either the original, if it was a failure, or the adjusted (imputed), if it was censored, or

$$y_i = t'_i\delta_i + (1 - \delta_i)\, t_i^+ = t'_i\delta_i + \delta'_i t_i^+,$$

and $y_1, y_2, \ldots, y_n$ be the complete adjusted data. For an adjustment of the censored value t'_i we can use the average of the already adjusted values *after* t'_i, or

$$y_i = t_i^+ = \frac{1}{n-i}\sum_{j=i+1}^{n} y_j,$$

as a simple and intuitively appealing method. Then, our estimator of the population mean will be

$$\hat{\mu} = y_0 = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

If all observation are failures, then simply $\hat{\mu} = y_0 = \frac{1}{n}\sum_{i=1}^{n} t'_j = \bar{t}$. The order of imputed censored values is preserved: if $t'_i \le t'_j$ then $t_i^+ \le t_j^+$. The calculation starts with y_n, which is not defined by our formula if t'_n is censored. Now, this is a more complex problem. We can use some special adjustment—correction to t'_n, such as adding a weighted difference $t'_n - t'_{n-1}$, or a simulated value from an exponential distribution (Brown, Hollander, and Kornwar 1974), which can be justified by the asymptotic distribution of the RUL (Banjevic 2009), or the extreme value theory (Alvarez-Iglesias et al. 2015), or often in engineering using an expert's opinion. Some confusion may arise if the last censored value is the same as the last failed value (which is placed *before* the censored values). It that case, only the last censored value needs to be adjusted, and the previous failed and censored values remain the same. To make it easier for analysis, we will use the simple and popular Efron's tail correction (Efron 1967) and define $y_n = t'_n$, and $\delta_n = 1$ regardless of the actual δ_n (in that case all values tied to t'_n will also be declared as failures). More on the right tail correction see in Brown, Hollander, and Kornwar (1974), Malla and Mukerjee (2010), and Khan and Shaw (2016a,b). The reviewer gave a suggestion: "However, because the correction closes the right tail when the largest observation is originally censored, it would be helpful to state explicitly what quantity is being estimated under this convention." A short discussion on the reviewer's suggestion is included in Note 16 (Section 9).

Note 1. It is easy to see that if t'_i is a suspension, then $t_{i-1}^+ = t_i^+$, and $t_i^+ = y_i$, regardless of whether t'_{i-1} is a suspension, or a failure. And, if t'_i is a failure, then $t_{i-1}^+ < t_i^+$, except for the failures tided to t'_n, when $t_{i-1}^+ = t_i^+ = \ldots = t'_n$. Assume that t'_i is a suspension. Then $\sum_{j=i+1}^{n} y_j = (n-i)\,y_i$, and $t_{i-1}^+ = \frac{1}{n-(i-1)}\sum_{j=i}^{n} y_j = \frac{1}{n-i+1}(y_i + \sum_{j=i+1}^{n} y_j) = \frac{1}{n-i+1}\left(y_i + (n-i)\,y_i\right) = y_i = t_i^+$. A proof for the second case, when t'_i is a failure, is left to the reader. If t'_{i-1} is also a suspension, then $y_{i-1} = y_i$, and then all suspensions between two consecutive failures have the same adjustment, regardless of their actual values. See also Example 1. There were some attempts to resolve this problem, to use actual suspended values, and we will mention them in Note 3. Still, we must note that there is no generally accepted solution to this problem.

It is of interest to express $\bar{y}$ in terms of the original values, that is, to find contribution ("weight") a_i of every t'_i to the average $\bar{y}$. Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{n}\sum_{i=1}^{n} a_i t'_i$. It is obvious from our construction that t'_i contributes to the calculation of $y_j, j \le i$, and not to $y_j, j > i$, so that a_i depends only on the values up to t'_i, not on the values after t'_i. Assuming $a_j, j < i$, have been found, write $n\bar{y} = \sum_{j=1}^{i-1} a_j t'_j + b_i \sum_{j=i}^{n} y_j$, where b_i a temporary "weight" of all remaining observations, with $b_1 = 1$. Then

$$b_i \sum_{j=i}^{n} y_j = b_i\left(\delta_i t'_i + \delta'_i \frac{1}{n-i}\sum_{j=i+1}^{n} y_j + \sum_{j=i+1}^{n} y_j\right)$$
$$= b_i\delta_i t'_i + \left(\frac{\delta'_i}{n-i} + 1\right) b_i \sum_{j=i+1}^{n} y_j,$$

or $a_i = \delta_i b_i$, and $b_{i+1} = b_i \frac{n-i+\delta'_i}{n-i} = b_i\left(\frac{n-i+1}{n-i}\right)^{1-\delta_i}$, as t'_i does not contribute anything to $\sum_{j=i+1}^{n} y_j$. Simply, if observation i is a failure, $a_i = b_i$, and $b_{i+1} = b_i$. In total

$$b_1 = 1, b_i = b_{i-1}\frac{n-i+1+\delta'_{i-1}}{n-i+1}$$
$$= b_{i-1}\left(\frac{n-i+2}{n-i+1}\right)^{1-\delta_{i-1}}, i = 2, 3, \ldots, n, \qquad (1)$$

and

$$a_i = \delta_i b_i = \delta_i \prod_{1\le j<i} \frac{n-j+\delta'_j}{n-j}$$
$$= \delta_i \prod_{1\le j<i}\left(\frac{n-j+1}{n-j}\right)^{1-\delta_j}, i = 1, 2, \ldots, n. \qquad (2)$$

Obviously, only failures contribute directly to the calculation, with appropriate weights. Censored values contribute through their positions between failures. But this is not all! If we consider $t_i^+ = \frac{1}{n-i}\sum_{j=i+1}^{n} y_j = \frac{1}{n-i}\sum_{j=i+1}^{n} a_j(i) t'_j$, with $\hat{\mu} = t_0^+$, we immediately get $a_j(i) = \delta_j b_j(i) = \delta_j \prod_{i+1\le l<j}\left(\frac{n-l+1}{n-l}\right)^{1-\delta_l} = \delta_j \frac{b_j}{b_{i+1}}$, if we apply our analysis to the sequence $t'_{i+1} \le t'_{i+2} \le \cdots \le t'_n$ Then, b_{i+1} is the average weight of $a_{i+1}, \ldots, a_n$, or

$$b_{i+1} = \frac{1}{n-i}\sum_{j=i+1}^{n} a_j = \frac{1}{n-i}\sum_{j=i+1}^{n} \delta_j b_j.$$

It is also easy to show that

$$t_i^+ = \frac{\delta_{i+1}}{n-i} t'_{i+1} + \left(1 - \frac{\delta_{i+1}}{n-i}\right) t_{i+1}^+, \text{ with } t_n^+ = t'_n,$$

but a more efficient recursive relation for the calculation of $\hat{\mu} = t_0^+$ will be introduced in Section 3. t_i^+ is, simply, the average life of the units failing at, or surviving t'_{i+1} (surviving *prior to* t'_{i+1}), with adjustments.

Note 2. From $b_1 = 1$ it follows immediately that $\frac{1}{n}\sum_{i=1}^{n} a_i = 1$, or $\sum_{i=1}^{n} a_i = n$, which means that the weights are "redistributed" from all values to the uncensored values only. This result also follows from Efron's 1967 "reweighting" idea (see below). It is also clear that the weights of consecutive failures are non-decreasing (because $b_1 \le b_2 \le \ldots \le b_n = a_n$), so the later failures contribute more to the estimation of the mean, due to censoring (they are included in all adjustments going backward).

The method of imputation with averages for the calculation of $\hat{\mu}$ produces the same weights as the "reweighting" that is mentioned in passing by Efron (1967, eq. (7.19)). Initially,

Table 1. Calculation of the mean estimator using recursion.

i	t'_i	δ_i	$n-i$	y_i	b_i	weights a_i	t_i^+
20	107	1	0	107	2.6169	2.6169	107
19	107	1	1	107	2.6169	2.6169	107
18	104	0	2	107	1.7446	0	107
17	97	1	3	97	1.7446	1.7446	107
16	97	1	4	97	1.7446	1.7446	104.5000
15	93	1	5	93	1.7446	1.7446	103.0000
14	75	1	6	75	1.7446	1.7446	101.3333
13	59	1	7	59	1.7446	1.7446	97.5714
12	54	0	8	92.7500	1.5508	0	92.7500
11	54	0	9	92.7500	1.3957	0	92.7500
10	38	0	10	92.7500	1.2688	0	92.7500
9	36	1	11	36	1.2688	1.2688	92.7500
8	36	1	12	36	1.2688	1.2688	88.0208
7	30	1	13	30	1.2688	1.2688	84.0192
6	23	0	14	80.1607	1.1842	0	80.1607
5	19	1	15	19	1.1842	1.1842	80.1607
4	18	0	16	76.3382	1.1146	0	76.3382
3	13	0	17	76.3382	1.0526	0	76.3382
2	10	1	18	10	1.0526	1.0526	76.3382
1	6	0	19	72.8467	1	0	72.8467
0	0	0	20	72.8467		$\Sigma a_i = 20$	72.8467

every observation t'_i gets weight $1/n$. Then the weight of the first censored value is distributed equally to *all* observations above it. Then the new weight of the second censored value is distributed equally to all observation above it, and so on. The final weights are equal to our weights a_i/n, redistributed to the uncensored values only. The redistribution of the weights, to the uncensored values on the right, can start from the largest top censored value (Dinse 1985). This is essentially the same method of creating weights as in our method of imputation that start from the right, but from completely different reason. The idea of "imputation" with averages that adjusts censored values comes as simple and intuitive, while an ad-hoc "redistribution of the weights" gives somewhat unclear "importance" to the failure values.

As it is mentioned in Section 1, Datta (2005) describes an estimator of the population mean (following Robins and Rotnitzky 1992) directly using weighted failure times, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} w_i t'_i$, where $w_i = 0$ if $\delta_i = 0$, and w_i is the inverse of the K-M estimator for the survivor function (see Section 4) of the *censored observations only*, $w_i = 1/\hat{S}_c(t'_i)$. This method also reduces to our imputation. Clearly, in our case $a_i = 0$ if $\delta_i = 0$, also. Let for $\delta_i = 1$,$e_i = 1/a_i = \prod_{1\le j<i} \frac{n-j}{n-j+\delta'_j} = \prod_{1\le j<i}\left(\frac{n-j}{n-j+1}\right)^{\delta'_j} = \prod_{1\le j\le i}\left(\frac{n-j}{n-j+1}\right)^{\delta'_j}$. Then exactly $\hat{S}_c(t'_i)$ at failure times, having in mind that δ'_i are indicators of censored times. This out-of-the-box solution is quite unintuitive.

Example 1. The data in the example is borrowed from Gillespie et al. (2010, Example 1.1, p. 5), being times in weeks to discontinuation of the use of an IUD (intrauterine device) (DUIUD in short), see Collett for more details. The data presented here is slightly altered to better meet our demonstration purposes (two records are also added to the original 18). We give the sample with censoring information in descending order to better illustrate calculation that starts from the top. In the original data, the last value, 107, is censored, but here it is declared as a failure, by default. The censoring rate is high, as 8 out of 20 values are incomplete. All calculations in this and the following examples are rounded to 4 decimal places. Notice that $n = 20, y_{20} = t'_{20} = 107, y_{19} = t'_{19} = 107, y_{18} = \frac{1}{2}(y_{19} + y_{20}) = 107$, and so on. $t'_1 = 6$ is censored, so that $y_1 = \frac{1}{19}\sum_{i=2}^{20} y_i = 72.8467$, and $\hat{\mu} = \bar{y} = y_0 = y_1 = \frac{1}{20}\sum_{i=1}^{20} y_i = 72.8467$ is the estimated mean time in weeks to DUIUD. $\bar{y} = y_0 = y_1$ because t'_1 is censored. See Table 1. Compare y_i and t_i^+. For example, $y_6 = t_6^+ = 80.1607$, because it is a suspension, but $y_7 = t'_7 = 30$, because it is a failure.

This method of recursive calculation is simple and less sensitive to rounding than using the Datta's formula $\bar{x}^* = \frac{1}{n}\sum_{i=1}^{n} \frac{x_i}{\hat{S}_C(x_i)}\delta_i$. The same result will be obtained if we ignore suspended values and use their counts only in a compressed table, as we will see in Section 3.

Note 3. It would be interesting if we can somehow utilize actual suspended values, instead of their positions only, but it would likely require some data model or interpolation between points (see, e.g., O'Quigley 2008, p. 142). In this paper we are primarily interested in nonparametric estimation. Essentially, the noninclusion of censored values comes as if we use nonparametric maximum likelihood.

3. Recursive Calculation of the Mean Estimator

Let $0 < t_1 < t_2 < \cdots < t_k$ are all *different* positive *failure* times, $k \le n$, including $t_k = t'_n$, that can be censored in the original data (or, even, both failed and censored), and let $t_0 = 0$. Let

$d_i = \#\{t'_l : t'_l = t_i, t'_l \text{ a failure}\}$—the number of ties (failures) at t_i, $d_i \ge 1, i \ge 1, d_0 \ge 0$ (failures at 0 are usually excluded, since they normally do not come from the same population, particularly in engineering, but here we allow them for convenience),

$c_i = \#\{t'_l : t'_l \text{ a suspension}, t_i \le t'_l < t_{i+1}\}, c_i \ge 0, 0 \le i < k, c_k = 0$($c_0$ counts suspension inside (t_0, t_1), no suspensions at t_0; all data at t_k is declared as failures),

$Y_i = \#\{t'_l : t_i \le t'_l\}$—the number of lifetimes at risk at t_i, including failures and suspensions at t_i, $Y_0 = n$, $Y_i = d_i + c_i + Y_{i+1}$, or $Y_i = Y_{i-1} - d_{i-1} - c_{i-1}$, $Y_k = d_k$ (all data at t_k is declared as failures).

Let the "t^*" sequence be defined by

$$t_i^* = \frac{t_i d_i + (Y_i - d_i)t_{i+1}^*}{Y_i} = t_i\frac{d_i}{Y_i} + (1 - \frac{d_i}{Y_i})t_{i+1}^* = t_i\hat{q}_i + t_{i+1}^*\hat{p}_i, \qquad (3)$$

where we use the notation $\hat{p}_i = 1 - \frac{d_i}{Y_i}, \hat{q}_i = \frac{d_i}{Y_i}, i = 1, 2, \ldots, k$. The calculation starts with $t_k^* = t_k\hat{q}_k + t_{k+1}^*\hat{p}_k = t_k$, where $\hat{q}_k = 1$ (from $Y_k = d_k$), and t_{k+1}^* is arbitrary. Notice that $t_0^* = t_1^*$ if $d_0 = 0$. Then for all censored observations t'_l,

$$y_l = t_i^*, t_{i-1} \le t'_l < t_i, \qquad (4)$$

or in the notation from Section 2, $y_l = t_i^+ = t_i^*$, and finally

$$\hat{\mu} = y_0 = t_0^*. \qquad (5)$$

See Proof 1. So, the calculation of y_l reduces to the recursive calculation of $t_i^*, i = 1, 2, \ldots, k$, in (3). Also, it is easy to show that

$$t_i^* - t_i = \left(t_{i+1}^* - t_i^*\right)\frac{\hat{p}_i}{\hat{q}_i} = \left(t_{i+1}^* - t_i^*\right)\frac{Y_i - d_i}{d_i}, \qquad (6)$$

which we will need later. In addition, we have that $t_i^* < t_{i+1}^*$, except $t_0^* \le t_1^*$, due to $d_0 \ge 0$, and we have that $t_i < t_i^*$, except for $t_k = t_k^*$, due to $d_k = Y_k$. See also Proof 1.

Note 4. What is t_i^*? It is the total (estimated) expected life if we survive prior to t_i (survived t_{i-1}). We still can fail at t_i with the probability $\hat{q}_i$ or we expect to live t_{i+1}^* with the probability $\hat{p}_i$ A notion of "life expectancy function" is discussed in actuarial studies. See also "vitality function," in Kupka and Loo (1989).

Example 2. We compress the data from Example 1 using failures and the counts of failures and suspensions only. Here we have $k = 9$, $Y_0 = 20$. We start with $t_9^* = t_9 = 107$, $t_8^* = t_8\frac{2}{5} + t_9^*\frac{3}{5} = 97\frac{2}{5} + 107\frac{3}{5} = 103$, and so on. $\bar{t} = t_0^* = t_1^* = 72.8467$ is the estimated expected DUIUD $\hat{\mu}$ for the subjects who just started using it. For example, $t_3^* = 80.1607$ is the estimated expected DUIUD for a subject that survived up to age $t_3 = 30$ (but still can fail—stop to use IUD, at time 30). If the subject *survived* age $t_3 = 30$, its estimated MRL is $t_4^* - t_3 = 84.0192 - 30 = 54.0192$ weeks. The last column shows contributions of the subjects in intervals $[t_i, t_{i+1})$, where every failure contributes t_i and suspension t_{i+1}^*. The average of all contributions (divided by Y_0) is $\bar{t} = t_0 = \hat{\mu}$ (Table 2).

4. K-M Estimator of the Mean

The function $\hat{S}(t) = \prod_{0 \le t_i \le t}(1 - \frac{d_i}{Y_i}) = \prod_{0 \le t_i \le t}\hat{p}_i = \hat{P}(T > t)$ is known as the K-M estimator of the survival function (the K-M paper; see also Meier (1975)). The formula comes from a simple idea that $\hat{S}(t_i) = \hat{P}(T > t_i) = \hat{P}(T > t_{i-1})\hat{P}(T > t_i | T > t_{i-1}) = \hat{S}(t_{i-1})\hat{P}(T > t_i | T \ge t_i) = \hat{S}(t_{i-1})\hat{p}_i$. Another approach to derive the K-M estimator is to use weights assigned in Section 2 to observed values $0 \le t'_1 \le t'_2 \le \cdots \le t'_n$ to define the probability mass function $\hat{P}(T = t'_i) = \frac{a_i}{n}$ (the K-M weight function, Mauro 1985), and then to calculate $\hat{S}'(t'_i-) = \hat{P}(T \ge t'_i) = \sum_{j \ge i}\hat{P}(T = t'_j) = \prod_{1 \le j < i}\left(\frac{n-j}{n-j+1}\right)^{\delta_j}$ (Efron 1967). Then $\hat{S}(t_i) = \hat{P}(T > t_i) = \hat{S}'(t_{i+1}-)$ at uncensored points. Fine details are left to the reader. Keep in mind that the weight of zero is assigned to censored points. So, the K-M estimator comes as a natural method to define empirical PMF from adjusted data, consistent with the weights obtained by imputation ($\hat{P}(T = t'_i) = \hat{P}(T > t'_{i-1}) - \hat{P}(T > t'_i) = \hat{S}(t'_{i-1}) - \hat{S}(t'_i)$). Another method to derive the K-M estimator of the survival function is through "self-consistency," also from Efron (1967). An equivalent introduction of self-consistency is given by Fisher and Kanarek (1974).

Note 5. If the failures are not tied, it is easy to see that $\hat{P}(T = t_i) = \hat{S}(t_{i-1}) - \hat{S}(t_i) \le S(t_i) - \hat{S}(t_{i+1}) = \hat{P}(T = t_{i+1})$ (see also Malla and Mukerjee 2010).

Note 6. It would be tempting to estimate $S(t)$ using the adjusted sample $y_1, y_2, \ldots, y_n$ as a complete sample of failure times, when it would be $\hat{S}'(t) = \frac{1}{n}\sum_{i=1}^{n} I\{y_i > t\}$, but we will not discuss it here. Obviously, $\hat{S}'(t)$ and the K-M estimator would be different, as they have different support ($\hat{S}'(t)$ would have larger support, depending on censoring).

Some basic properties of $\hat{S}(t)$ that will be needed in our analysis are:

- By definition $\hat{S}(t_{-1}) = \hat{S}(-0) = 1$, $\hat{P}(T = 0) = \hat{q}_0 \ge 0$, $\hat{S}(0) = \hat{P}(T > 0) = \hat{p}_0 = 1 - \hat{q}_0 \le 1$,
- $\hat{S}(t_k) = 0$, $\hat{P}(T > t_i) = \hat{S}(t_i)$, $\hat{P}(T \ge t_i) = \hat{S}(t_i-) = \hat{S}(t_{i-1})$,
- $\hat{P}(T > t | T > t_{i-1}) = \hat{P}\left(T > t | \, T \ge t_i\right) = \hat{S}(t | t_{i-1}) = \frac{\hat{S}(t)}{\hat{S}(t_{i-1})}, t \ge t_{i-1}$.
- $\hat{S}(t)$ is a step function, right continuous, with left limits and jumps down of size $\hat{P}(T = t_i) = \hat{S}(t_{i-1}) - \hat{S}(t_i) = \hat{S}(t_{i-1})\hat{q}_i$.

Table 2. Calculation of the mean estimator reduced to failure times.

i	t_i	d_i	c_i	$d_i + c_i$	Y_i	$Y_i - d_i$	t_i^*	$d_i t_i + c_i t_{i+1}^*$
9	107	2	0	2	2	0	107	214
8	97	2	1	3	5	3	103	301
7	93	1	0	1	6	5	101.3333	93
6	75	1	0	1	7	6	97.5714	75
5	59	1	0	1	8	7	92.7500	59
4	36	2	3	5	13	11	84.0192	350.25
3	30	1	0	1	14	13	80.1607	30
2	19	1	1	2	16	15	76.3382	99.1607
1	10	1	2	3	19	18	72.8467	162.6763
0	0	0	1	1	20	20	72.8467	72.8467
						Y_0	$\hat{\mu}$	$\hat{\mu} = \sum/20$

For an example, see Figure 1 that shows the estimated survival function $\hat{S}(t)$ for DUIUD data (see Table 3).

Let $\hat{\mu}' = \int_0^\infty \hat{S}(t)\,dt = \int_0^{t_k} \hat{S}(t)\,dt$. Consider the integral $\int_{t_i}^{t_k} \hat{S}(t)\,dt = \sum_{j=i}^{k-1} \hat{S}(t_j)\left(t_{j+1} - t_j\right)$. After a couple of steps, we can see that

$$\int_{t_i}^{t_k} \hat{S}(t)\,dt = \sum_{j=i}^{k} [\hat{S}(t_{j-1}) - \hat{S}(t_j)]t_j - \hat{S}(t_{i-1})\,t_i$$
$$= \sum_{j=i}^{k} t_j \hat{S}\left(t_{j-1}\right) q_j - \hat{S}(t_{i-1})\,t_i,$$

and that $\hat{\mu}' = \sum_{i=1}^{k} [\hat{S}(t_{i-1}) - \hat{S}(t_i)]t_i = \sum_{i=1}^{k} t_i \hat{S}(t_{i-1})\hat{q}_i$.

Let $\hat{\mu}_i = \hat{E}(T|T \geq t_i)$, so that $\hat{\mu}' = \hat{\mu} = \hat{\mu}_0 = t_0^*$ (Proof 2). Then $\hat{\mu}_i = t_i^* = t_i + \hat{E}(T - t_i|T \geq t_i) = t_i + \int_{t_i}^{t_k} \hat{S}(t|T \geq t_i) = t_i + \int_{t_i}^{t_k} \hat{S}(t|t_{i-1})dt = t_i + \frac{1}{\hat{S}(t_{i-1})} \int_{t_i}^{t_k} \hat{S}(t)dt$, and then

$$\int_{t_i}^{t_k} \hat{S}(t)dt = \left(t_i^* - t_i\right)\hat{S}(t_{i-1}) = \left(t_{i+1}^* - t_i\right)\hat{S}(t_i)$$
$$= \left(t_{i+1}^* - t_i^*\right)\frac{\hat{S}(t_i)}{\hat{q}_i}. \tag{7}$$

See Note 4 about the meaning of $\hat{\mu}_i = t_i^*$ equation (6), and Proof 2. Similarly, let the MRL *after* t_i (if we survive t_i) be $m(t_i) = E(T - t_i|T > t_i)$. Then from (7),

$$\hat{m}(t_i) = \int_{t_i}^{t_k} \hat{S}(t|T > t_i)\,dt = t_{i+1}^* - t_i.$$

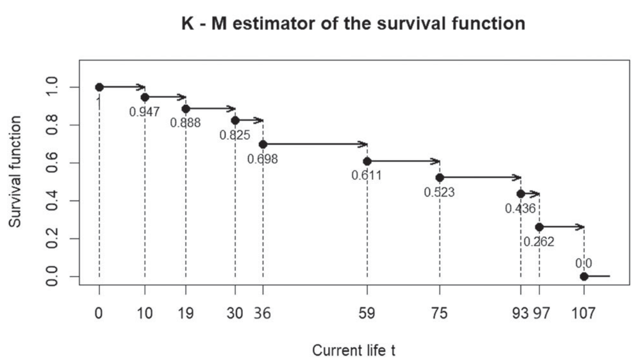

Figure 1. Survival function $\hat{S}(t)$ for DUIUD data.

Note 7. In this paper we use somewhat different definition of MRL, $\tilde{m}(t) = E(T - t|T \geq t)$ (we may fail at t), so that $\widehat{\tilde{m}}(t_i) = \hat{E}(T - t_i|T \geq t_i) = \int_{t_i}^{t_k} \hat{S}(t|T \geq t_i)dt = t_i^* - t_i$. Another formula for $\widehat{\tilde{m}}(t) = \hat{E}(T - t|T \geq t)$ can be found in Qin and Zhao (2007).

Note 8. Datta (2005) proves that the sample mean of the "sample" in which censored values t_i' are imputed with $\hat{E}\left(T|T > t_i'\right)$ coincides with the proper K-M estimator $\hat{\mu}'$. This method is equivalent to our method that uses the sequence $y_1, y_2, \ldots, y_n$ Datta's proof is based on the self consistency equation of the K-M estimator of the survival curve (Efron 1967), which is neither intuitive nor elementary. He also does not use any recursive relationships in his method. Our approach reduces the calculation of $\hat{\mu}'$ to the simple recursive computation of the sequence $t_0^*, t_1^*, \ldots, t_k^*$, without need for the whole sequence.

5. Variance Estimation

We also want to estimate σ^2 from censored data. It seems this problem is of less interest in biomedical studies (at least, see Royston and Parmar 2013), but is of more interest in engineering. A "naïve" estimator would be the sample variance $S'^2 = \frac{1}{n-1}\sum_i \left(t_i' - \overline{t'}\right)^2$ but it would be biased due to censoring. Less naïve approach would be to use our adjusted data y_i instead, or to calculate $S^2 = \frac{1}{n}\sum_i (y_i - \hat{\mu})^2 = \frac{1}{n}\sum_i (y_i - t_0^*)^2$ (we use division by n for simplicity). It might be expected that this estimator is also biased (but much less than S'^2), as the adjustment reduces variation, depending on the amount of censoring; we will discuss this problem later. We will consider one more approach in which we will use an adjustment for the squares, $\left(t_i^2\right)^*$, by "fitting" t_i^2 directly (essentially a "K-M" type estimator, as we shall see). Anyway, in both cases we want to reduce our calculation to t_i^* sequence. Let the estimated conditional variance for the units surviving to t_i (or the estimated variance of the MRL $t_i^* - t_i$) be

$$S_i^{*2} = \hat{V}ar(T - t_i|T \geq t_i) = \frac{1}{Y_i} \sum_{j \geq i: l, t_j \leq t_l' < t_{j+1}} y_l^2 - t_i^{*2}$$
$$= \frac{1}{Y_i} \sum_{j=i}^{k} \left(t_j^2 d_j + t_{j+1}^{*2} c_j\right) - t_i^{*2}.$$

Then $\hat{\sigma}^2 = S_0^{*2} = \hat{V}ar(T) = \hat{V}ar(T|T \geq 0)$ where $Y_0 = n$, is our estimator of the population variance.

Table 3. Calculating estimators of the variance.

t_i	d_i	c_i	Y_i	t_i^*	$S(t_i)$	S_i^{*2}	V_i	s_i^2	v_i^2
0	0	1	20	72.8467	1.0000	896.7652	1127.8672	0.0000	0.0000
10	1	2	19	72.8467	0.9474	943.9633	1127.8672	208.4567	219.4281
19	1	1	16	76.3382	0.8882	860.3856	958.9080	175.3422	207.6420
30	1	0	14	80.1607	0.8247	732.8090	789.0456	135.4822	171.8994
36	2	3	13	84.0192	0.6978	580.7448	641.3073	272.5091	345.7588
59	1	0	8	92.7500	0.6106	262.4375	262.4375	65.0893	113.5545
75	1	0	7	97.5714	0.5234	113.9592	113.9592	29.7190	51.8477
93	1	0	6	101.3333	0.4361	33.8889	33.8889	4.1667	7.2691
97	2	1	5	103.0000	0.2617	24.0000	24.0000	6.0000	10.4676
107	2	0	2	107.0000	0.0000	0.0000	0.0000	0.0000	0.0000
						896.7652	1127.8672	896.7652	1127.8672
						S_0^{*2}	V_0	$S_0^{*2} = \sum_{i=0}^{k} s_i^2$	$V_0 = \sum_{i=0}^{k} v_i^2$
								$S_0^* = 29.9460$	$\sqrt{V_0} = 33.5837$

We can derive a simple recursive formula for S_i^{*2},

$$S_i^{*2} = (t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{\hat{q}_i} + \frac{Y_{i+1}}{Y_i} S_{i+1}^{*2} = \frac{1}{Y_i} \sum_{j=i}^{k} (t_{j+1}^* - t_j^*)^2 \frac{\hat{p}_j}{\hat{q}_j} Y_j$$
$$= \frac{1}{Y_i} \sum_{j=i}^{k} (t_{j+1}^* - t_j^*)^2 \frac{Y_j \left(Y_j - d_j\right)}{d_j}, \tag{8}$$

with

$$S_0^{*2} = \frac{1}{Y_0} \sum_{i=0}^{k} (t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{\hat{q}_i} Y_i. \tag{9}$$

For $d_0 = 0$, $\hat{q}_0 = 0$, but also $t_1^* - t_0^* = 0$, and we assume, as usual, $\frac{0}{0} = 0$. See Proof 3.

For the other more theoretically structured approach, we need the Riemann-Stieltjes (R-S) integral, of which a short overview is given in Appendix C. Working with R-S integrals can be a tricky business due to discontinuous functions that appear in integration. We start with a general formula for a K-M type integral of a continuous function $g(x)$, $\int g(x) d\hat{F}(x) = \sum_i \int_{t_{i-1}}^{t_i} g(x) d\hat{F}(x) = \sum_i g(t_i)[\hat{S}(t_{i-1}) - \hat{S}(t_i)] = \sum_i g(t_i)\hat{S}(t_{i-1})\hat{q}_i$. Then a K-M type estimator for σ^2 is

$$\hat{\sigma}^2 = V_0 = \int (x - \hat{\mu})^2 d\hat{F}(x) = \int x^2 d\hat{F}(x) - \hat{\mu}^2$$
$$= \sum_{i=0}^{k} t_i^2 \hat{S}(t_{i-1}) \hat{q}_i - t_0^{*2},$$

as we already know that $\mu = t_0^* = \int x d\hat{F}(x)$. Let

$$V_i = \sum_{j=i}^{k} t_j^2 \hat{S}(t_{j-1}|t_{i-1}) \hat{q}_j - t_i^{*2}$$

be an appropriate K-M estimator of the variance of the MRL $t_i^* - t_i$. If we start with $V_k = t_k^2 \hat{S}(t_{k-1}|t_{k-1}) q_k - t_k^{*2} = t_k^2 - t_k^2 = 0 (\hat{q}_k = 1)$, we can show that

$$V_i = (t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{\hat{q}_i} + \hat{p}_i V_{i+1} = \sum_{j=i}^{k} (t_{j+1}^* - t_j^*)^2 \hat{S}(t_{j-1}|t_{i-1}) \frac{\hat{p}_j}{\hat{q}_j}$$
$$= \sum_{j=i}^{k} (t_{j+1}^* - t_j^*)^2 \hat{S}(t_j|t_{i-1}) \frac{1}{\hat{q}_j}, \tag{10}$$

so that

$$V_0 = \sum_{i=0}^{k} (t_{i+1}^* - t_i^*)^2 \hat{S}(t_i) \frac{1}{\hat{q}_i} = \sum_{i=0}^{k} (t_{i+1}^* - t_i^*)^2 \hat{S}(t_{i-1}) \frac{\hat{p}_i}{\hat{q}_i}. \tag{11}$$

See Proof 4.

Note 9. If there is no censoring, V_0 reduces to the sample variance $S_t^2 = \frac{1}{n} \sum_{i=1}^{n} (t_i' - \bar{t'})^2$ with divisor n. To see it, use that $\hat{S}(t_{i-1}) \frac{d_i}{Y_i} = \prod_{j=0}^{i-1} \frac{Y_j - d_j}{Y_j} \times \frac{d_i}{Y_i} = \frac{Y_{i-1} - d_{i-1}}{Y_0} \times \frac{d_i}{Y_i} = \frac{d_i}{Y_0}$, as $Y_j = Y_{j-1} - d_{j-1}$ $(c_{j-1} = 0)$, and then $V_0 = \sum_{j=0}^{k} t_j^2 \hat{S}(t_{j-1}) \frac{d_j}{Y_j} - t_0^{*2} = \frac{1}{Y_0} \sum_{j=1}^{k} t_j^2 d_j - \bar{t'}^2 = \frac{1}{n} \sum_{i=1}^{n} t_i'^2 - \bar{t'}^2 = S_t^2$.

Estimators S_0^{*2} and V_0 are different because we use different methods to estimate $E(T^2)$. Squaring increases variations, so using the *squares of imputed lifetimes* t_{i+1}^* in $\frac{1}{Y_0} \sum_i (t_i^2 d_i + (t_{i+1}^*)^2 c_i)$ for the calculation of S_0^{*2}, is different and should be less accurate that directly *imputing the squares of the lifetimes* $(t_{i+1}^2)^*$ in $\frac{1}{Y_0} \sum_i [t_i^2 d_i + (t_{i+1}^2)^* c_i]$ for the calculation of V_0. Details are left in Proof 5. Simply speaking (as the reviewer commented), V_0 addresses this bias through appropriate weighting. For a more formal justification, and further discussion, see Note 12.

Note 10. We can see that $S_0^{*2} \leq V_0$ (and $S_i^{*2} \leq V_i$ for all i). Let $S_0^{*2} = \sum_{i=0}^{k} s_i^2, s_i^2 = (t_{i+1}^* - t_i^*)^2 \frac{Y_i \hat{p}_i}{Y_0 \hat{q}_i}$ (see (3)), and $V_0 = \sum_{i=0}^{k} v_i^2, v_i^2 = (t_{i+1}^* - t_i^*)^2 \hat{S}(t_i) \frac{1}{\hat{q}_i}$ (see (11)), where s_i^2 and v_i^2 are the "contributors to the variance" of each failure point. Then $v_i^2 - s_i^2 = (t_{i+1}^* - t_i^*)^2 \frac{1}{\hat{q}_i} \left(\hat{S}(t_i) - \frac{Y_i \hat{p}_i}{Y_0} \right) = (t_{i+1}^* - t_i^*)^2 \frac{Y_i \hat{p}_i}{\hat{q}_i} (\frac{Y_0}{Y_i} \hat{S}(t_{i-1}) - 1) \geq 0$ for all i, as, simply, $\hat{S}(t_j) \geq \frac{Y_j \hat{p}_j}{Y_0}$, or $\hat{S}(t_{j-1}) \geq \frac{Y_j}{Y_0}$ (notice that $Y_j - d_j \geq Y_j - d_j - c_j = Y_{j+1}$). Then also, $S_0^{*2} \leq V_0$.

Note 11. From Note 10 we can write $V_0 - S_0^{*2} = \frac{1}{Y_0} \sum_{j=0}^{k} (t_{j+1}^* - t_j^*)^2 \frac{Y_j \hat{p}_j}{\hat{q}_j} B_C(t_j)$, where $B_C(t_j) = \frac{Y_0}{Y_j} \hat{S}(t_{j-1}) - 1 = \prod_{0<l\leq j} (1 + \frac{c_{l-1}}{Y_l}) - 1$(from $Y_{l-1} - d_{l-1} = Y_l + c_{l-1}$), $B_C(0) = 0$. The function $B_C(t) = \prod_{0<t_l\leq t} (1 + \frac{c_{l-1}}{Y_l}) - 1$ with $B_C(t) \approx (\geq) \sum_{0<t_l\leq t} \frac{c_{l-1}}{Y_l}$ can be interpreted as a cumulative measure of censoring over the interval $[t_0, t_k]$, better than just % of censoring (compare it with a widely used measure of "cumulative hazard" (Klein and Moeschberger 2003). If there were no censoring, $B_C(t_j) = 0$, or $\hat{S}(t_{j-1}) = \hat{P}(T \geq t_i) = \frac{Y_j}{Y_0}$ exactly, as expected. The function $B_C(t_j)$ will appear again in Section 6. The approximation above comes from $\prod_{j=0}^{i} (1 + x_j) \approx 1 + \sum_{j=0}^{i} x_i$ for $x_j \geq 0$ and $\sum_{j=0}^{i} x_i$ small; in any case, $\prod_{j=0}^{i} (1 + x_j) \geq 1 + \sum_{j=0}^{i} x_j$.

Note 12. The recursive calculation works for any continuous function $g(x)$, that is, for $\int g(x) d\hat{F}(x) = \sum_i g(t_i)(\hat{S}(t_{i-1}) - \hat{S}(t_i)) = \sum_i g(t_i)\hat{S}(t_{i-1})\hat{q}_i$. If we define $g_k^* = g(t_k)$ and $g_i^* = g(t_i)\hat{q}_i + g_{i+1}^* \hat{p}_i$, then $g_0^* = \int g(x) d\hat{F}(x)$. Also, $g_0^* = \frac{1}{n} \sum_{i=1}^{n} a_i g(t_i')$, if we take that the K-M weights are $d\hat{F}(t_i') = \hat{F}(t_i') - \hat{F}(t_i'-) = \frac{a_i}{n}$. For the proofs and further comments, see Proof 6.

Example 3. Estimating the population variance. We follow the data and calculation from Example 2, but in increasing order. $\hat{\mu} = t_0^* = 72.8467 S_0^{*2} = 896.7652 V_0 = 1127.8672$ Even if $S_0^{*2} < V_0$ (see Note 10), and censoring is not moderate (8 out of 20 values censored), an effect on the standard deviation is not extreme in comparison with the mean, $S_0^* = 29.9460$, and $\sqrt{V_0} = 33.5837$. S_0^{*2} and V_0 are calculated using recursion with $S_i^{*2}$and V_i, and directly, using the sequence t_i^*; notice that $S_0^{*2} = \frac{Y_1}{Y_0} S_1^{*2} = \frac{19}{20} S_1^{*2}$ due to $\frac{0}{0} = 0$ convention. See Table 3. The standard deviation is relatively large, which is not surprising, with DUIUD relatively evenly spaced between 0 and 107.

It is also interesting to compare the contribution of every failure point to the estimation of the variances in S_0^{*2} and V_0, (see Note 10). The contributions fluctuate but are consistent with the size of censoring at that point. As proved, $s_i^2 \leq v_i^2$. Most of the contributions come from the values less than or equal $t_i = 36$. This makes sense because the earlier censored values are less accurately imputed by t_i^+ than the later ones.

6. Estimating Variance of the Sample Mean

We have discussed in detail the estimator $\hat{\mu}$ of the population mean, but $\hat{\mu}$ is of little value without an estimator of its accuracy, which is commonly measured by its variance $Var(\hat{\mu})$. In the case without censoring, we usually use $\hat{V}ar\left(\hat{\mu}\right) = \frac{\hat{\sigma}^2}{n}$, but it is not clear how it would work with censoring, because $\hat{\mu}$ is a complicated function of the sample, not just the simple sample mean. We will discuss this problem soon. We need another approach that accounts for censoring in the formula for $Var(\hat{\mu})$. Kaplan and Meier (1958) proposed their classical estimator for the variance of the sample mean $\hat{\mu} = \int_0^{t_k} \hat{S}(t)dt$

$$\hat{V}ar_0\left(\hat{\mu}\right) = \sum_{i=0}^{k} [\int_{t_i}^{t_k} \hat{S}\left(t\right)dt]^2 \frac{d_i}{Y_i\left(Y_i - d_i\right)}. \tag{12}$$

The estimator does not seem a natural or obvious first choice if one relies on elementary arguments. At first glance, it is hardly intuitive and appears somewhat mysterious. This was one of our reasons for examining the original paper to understand the logic behind it. The logic is fairly simple and depends on integrals of the type $\int_a^{t_k} \hat{S}(t)dt$ but requires some mathematics of integration, as we will see in Section 7. Hosmer and Lemeshow (1999, p. 55) have a similar comment: "*The estimator of the variance of the sample mean is neither particularly intuitive, nor easy to motivate so we just provide it and demonstrate the calculation.*"

From our work in Section 4, we can easily reduce the calculation of $\hat{V}ar_0(\hat{\mu})$ to a recursive calculation with the sequence t_i^*. From (6) and (7), $\int_{t_i}^{t_k} \hat{S}(t)dt = (t_i^* - t_i)\hat{S}(t_{i-1})$ and $t_i^* - t_i = (t_{i+1}^* - t_i^*)\frac{\hat{p}_i}{\hat{q}_i}$ (for $i = 0$, $\hat{S}\left(t_{-1}\right) = \hat{S}\left(-0\right) = 1$). Then

$$\begin{aligned}\hat{V}ar_0(\hat{\mu}) &= \sum_{i=0}^{k} \hat{S}^2(t_{i-1})(t_i^* - t_i)^2 \frac{\hat{q}_i}{Y_i\hat{p}_i} \\ &= \sum_{i=0}^{k} \hat{S}^2(t_{i-1})(t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{Y_i\hat{q}_i} \\ &= \sum_{i=0}^{k} \hat{S}^2(t_i)(t_{i+1}^* - t_i^*)^2 \frac{1}{Y_i\hat{q}_i\hat{p}_i}, \\ &= \sum_{i=0}^{k} \hat{S}^2(t_i)(t_{i+1}^* - t_i^*)^2 \frac{Y_i}{d_i(Y_i - d_i)}\end{aligned} \tag{13}$$

(for $i = 0$ we may have $\frac{0}{0} = 0$). For recursive calculation of $\hat{V}ar_0(\hat{\mu})$, let us first introduce the estimator of the conditional variance (see also Kumazawa 1987).

$$\begin{aligned}A_i &= \hat{V}ar_0\left(\hat{\mu}_i|T \geq t_i\right) = \hat{V}ar_0\left(\hat{\mu}_i|T > t_{i-1}\right) \\ &= \sum_{j=i}^{k} \hat{S}^2(t_{j-1}|t_{i-1})(t_{j+1}^* - t_j^*)^2 \frac{\hat{p}_j}{Y_j\hat{q}_j}, A_k = 0.\end{aligned}$$

Then $\hat{V}ar_0(\hat{\mu}) = A_0$. It is easy to get a simple recursive formula for A_i,

$$\begin{aligned}A_i &= (t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{Y_i\hat{q}_i} + p_i^2 \sum_{j=i+1}^{k} \hat{S}^2(t_{j-1}|t_i)(t_{j+1}^* - t_j^*)^2 \frac{\hat{p}_j}{Y_j\hat{q}_j} \\ &= (t_{i+1}^* - t_i^*)^2 \frac{\hat{p}_i}{Y_i\hat{q}_i} + \hat{p}_i^2 A_{i+1}.\end{aligned} \tag{14}$$

Note 13. If there were no censoring, $Var\left(\hat{\mu}\right) = \frac{\sigma^2}{n}$, so it would be interesting to compare an estimator for σ^2/n with an estimator for $Var\left(\hat{\mu}\right)$, both from censored data. Let $\hat{V}ar_0\left(\hat{\mu}\right) = \sum_{i=0}^{k} u_i^2$, where $u_i^2 = \hat{S}^2(t_i)(t_{i+1}^* - t_i^*)^2 \frac{1}{Y_i\hat{q}_i\hat{p}_i}$. From (13) we have that $u_i^2 - v_i^2/Y_0 = (t_{i+1}^* - t_i^*)^2 \left(\hat{S}^2\left(t_i\right) \frac{1}{Y_i\hat{q}_i\hat{p}_i} - S\left(t_i\right) \frac{1}{Y_0\hat{q}_i}\right) = (t_{i+1}^* - t_i^*)^2\hat{S}(t_i) \frac{1}{Y_0\hat{q}_i} B_C(t_i) \geq 0$ (see Note 11), and then $\hat{V}ar_0\left(\hat{\mu}\right) \geq V_0/n$. Altogether, with Note 10, $n\hat{V}ar_0\left(\hat{\mu}\right) \geq V_0 \geq S_0^{*2}$, with the "contributors" $Y_0u_i^2 \geq v_i^2 \geq s_i^2$. A numerical illustration of the comparison $Y_0u_i^2 \geq v_i^2$ is given in Example 4.

Note 14. If there is no censoring we have $B_C\left(t_i\right) = 0$, and $\hat{V}ar_0(\hat{\mu}) = \frac{1}{n}V_0 = \frac{1}{n}S_t^2$, a known result. With an increase of censoring the difference $\hat{V}ar_0\left(\hat{\mu}\right) - \frac{V_0}{n}$ will likely increase, so the ratio $0 \leq \frac{\hat{V}ar_0(\hat{\mu}) - V_0/n}{\hat{V}ar_0(\hat{\mu})} = R_c < 1$ may be useful as a relative measure of censoring. It seems that a more detailed analysis of the function $B_C(t_i)$ could be of interest.

Example 4. Estimating the variance of $\hat{\mu}$. We follow the data and calculation from Example 3. $\hat{\mu} = t_0^* = 72.8467\hat{V}ar_0\left(\hat{\mu}\right) = 72.6588$ is calculated using recursion and direct summation. $\frac{V_0}{20} = 56.3934$, calculated by summation (see Example 3), and is included for comparison with $\hat{V}ar_0(\hat{\mu})$. $\sqrt{\hat{V}ar_0(\hat{\mu})} = 8.5240$ is not significantly larger than $\sqrt{V_0/n} = 7.5096$.

The variance contributions of each point $v_i^2/20$ and u_i^2 fluctuate but are consistent with the size of censoring at that point and with its position, as discussed in Example 3. As proved, $s_i^2 \leq v_i^2 \leq nu_i^2$. If we use the proposed measure of censoring, we get $R_c = \frac{\hat{V}ar_0(\hat{\mu}) - V_0/n}{\hat{V}ar_0(\hat{\mu})} = \frac{72.6588 - 56.3934}{72.6588} = 22.39\%$, not very high, despite of relatively high level of censoring of 8/20 = 40%. The estimation is illustrated in Figure 2. We plot the increasing sequence t_i^*—the total expected life at the point t_i (solid line) and include the error of estimation as $\hat{\sigma}\left(t_i^*\right) = \sqrt{A_i} = \sqrt{\hat{V}ar_0(\hat{\mu}_i|T \geq t_i)}$ (two dotted lines, $t_i^* \pm \hat{\sigma}\left(t_i^*\right)$. For example, at the current life of $t_3 = 30$, the total expected life is $t_3^* = 80.16 \pm 7.87$, and at $t_5 = 59$, it is $t_5^* = 92.75 \pm 5.73$. Clearly, $\hat{\mu} = t_0^* = 72.85$ and $\hat{\sigma}\left(\hat{\mu}\right) = \hat{\sigma}\left(t_0^*\right) = \sqrt{\hat{V}ar_0(\hat{\mu})} = 8.52$. On the other side, $t_9^* = 107$, and $\hat{\sigma}\left(t_9^*\right) = 0$, since we reached the maximum life, and there is no more uncertainty in estimation.

7. A Theoretical Background for the K-M Estimator of the Variance; a Logic behind the Formula

The formula for $\hat{V}ar_0(\hat{\mu})$ was first introduced in the original K-M paper and was inspired by Greenwood's formula and Irwin's

Table 4. Calculating estimators of the variance of $\hat{\mu}$.

t_i	d_i	c_i	Y_i	t_i^*	$\hat{S}(t_i)$	A_i	$v_i^2/20$	u_i^2
0	0	1	20	72.8467	1.0000	72.6588	0.0000	0.0000
10	1	2	19	72.8467	0.9474	72.6588	10.9714	11.5488
19	1	1	16	76.3382	0.8882	68.0885	10.3821	12.2946
30	1	0	14	80.1607	0.8247	61.8836	8.5950	10.9053
36	2	3	13	84.0192	0.6978	55.7370	17.2879	21.9349
59	1	0	8	92.7500	0.6106	32.8047	5.6777	9.9053
75	1	0	7	97.5714	0.5234	16.2799	2.5924	4.5227
93	1	0	6	101.3333	0.4361	5.6481	0.3635	0.6341
97	2	1	5	103.0000	0.2617	4.8000	0.5234	0.9131
107	2	0	2	107.0000	0.0000	0.0000	0.0000	0.0000
						72.6588	56.3934	72.6588
						$\hat{V}ar_0\left(\hat{\mu}\right) = A_0$	$V_0/20 = \sum_{i=0}^{k} v_i^2/20$	$\hat{V}ar_0\left(\hat{\mu}\right) = \sum_{i=0}^{k} u_i^2$
							$\sqrt{\frac{V_0}{n}} = 7.5096$	$\sqrt{\hat{V}ar_0(\hat{\mu})} = 8.5240$

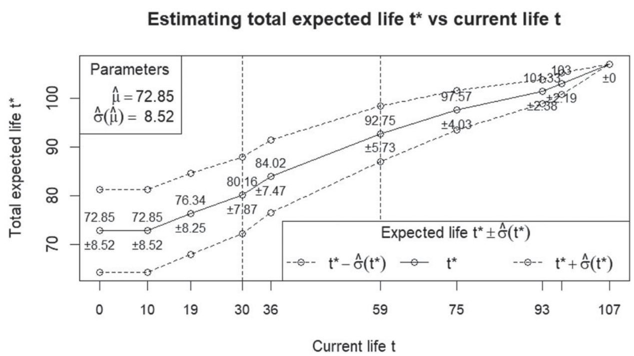

Figure 2. Estimating total expected life t_i^* and the error of estimation.

1949 approach, as the authors explain. It is interesting that their derivation uses simple arguments of conditional expectation, similar to what was later formalized and used in the counting processes theory. A simplified stochastic model for the data of our interest follows main ideas from the K-M paper and will help us to give a glimpse in the "mystery" of the formula, at least on the level of the model. Real derivation uses more general arguments, but similar in essence.

1. The points $t_0 = 0 < t_1 < t_2 < \cdots < t_k$ (intervals $[t_i, t_{i+1}), i = 0, 1, 2, \ldots$) are fixed in advance.
2. In interval i we start with a given number $N_i > 0$ of subjects (people, equipment) of age t_i and follow them prior to age t_{i+1}. We observe d_i deaths (failures) and c_i suspensions (censorings). Given N_i and t_i, d_i and c_i are conditionally independent of the past. For simplicity we assume that a subject can either die (fail) at t_i, or survive prior to t_{i+1}, but can be censored in mean time, which makes our model a simple discrete-jump model. It means that even if a subject is censored *before* t_{i+1}, it cannot die before t_{i+1}.
3. Let q_i be the probability of failing before t_{i+1}, and $p_i = 1 - q_i$ of surviving up to t_{i+1}, given survival (or age) up to t_i, or $q_i = P(T < t_{i+1}|T \geq t_i) = P(T = t_i|T \geq t_i)$ and $p_i = P(T \geq t_{i+1}|T \geq t_i) = P(T > t_i|T \geq t_i)$. $\hat{q}_i = \frac{d_i}{N_i}$ and $p_i = 1 - \frac{d_i}{N_i}$ are assumed unbiased estimators of q_i and p_i, or $E_{t_i-}(\hat{q}_i) = q_i$, $E_{t_i-}(\hat{p}_i) = p_i$.

Our conditions are not realistically met in typical followup studies in the biomedical and industrial fields, where the intervals $[t_i, t_{i+1})$ and the variables$N_i = Y_i > 0$ are random rather than fixed in advance. Still, conditioning on the data prior to t_i in that case is not unrealistic, due to assumed independent censoring. In that case $N_i = N_{i-1} - d_{i-1} - c_{i-1}$ is known, and d_i and c_i are independent of the past, given t_i and N_i A similar simplified model is used by Efron (1988), with a fixed grid t_i and N_i "*fixed at their observed values*" (see the paragraph after his eq. (2.3)). In the extensive literature, results on the expectation, bias, and convergence of various functionals of the K–M estimator are typically derived under the somewhat restrictive assumption of *random censorship*, in which the lifetime and its censoring time are independent random variables (see, e.g., Gather and Pawlitschko 1998).

We will use our recursive approach to establish the theoretical formula for $Var\left(\hat{\mu}\right)$ and then just introduce its sample analog. We start with the theoretical recursive formula for $\mu_i = E\left(T|T \geq t_i\right)$.

Proposition 1. Let $\mu_i = E\left(T|T \geq t_i\right)$. Under the stochastic model 1-3 introduced above

$$\mu_i = t_i q_i + \mu_{i+1} p_i, i = 0, 1, \ldots, k-1, \mu_k = t_k. \tag{15}$$

Proof. $\mu_i = E\left(T|T \geq t_i\right) = E\left(T|T = t_i, T \geq t_i\right) P\left(T = t_i|T \geq t_i\right) + E\left(T|T > t_i, T \geq t_i\right) P\left(T > t_i|T \geq t_i\right) = t_i P\left(T = t_i|T \geq t_i\right) + E\left(T|T \geq t_{i+1}\right) P\left(T > t_i|T \geq t_i\right) = t_i q_i + \mu_{i+1} p_i$. Notice that $\mu_k = E\left(T|T \geq t_k\right) = t_k$. □

The sample analog of (15) is our familiar formula $\hat{\mu}_i = t_i^* = t_i \hat{q}_i + \hat{\mu}_{i+1} \hat{p}_i = t_i \hat{q}_i + t_{i+1}^* \hat{p}_i$. We even can just start from it, without using Proposition 1, but Proposition 1 gives us a good starting point for our recursive calculation. So, we are interested in finding $Var\left(\hat{\mu}\right) = Var(\hat{\mu}_0)$. A slightly more convenient will be to work with $m_i = E\left(T - t_i|T \geq t_i\right) = E\left(T|T \geq t_i\right) - t_i = \mu_i - t_i = p_i(\Delta t_{i+1} + m_{i+1})$ (from (15)) and its sample analog $\hat{m}_i = \hat{p}_i(\Delta t_{i+1} + \hat{m}_{i+1})$. Then $Var\left(\hat{m}_i\right) = Var\left(\hat{\mu}_i - t_i\right) = Var\left(\hat{\mu}_i\right)$.

Proposition 2. Let $c_i = Var\left(\hat{\mu}_i\right) = Var\left(\hat{m}_i\right)$. Under the stochastic model 1-3 introduced above

$$c_i = \left(\mu_{i+1} - t_i\right)^2 Var\left(\hat{p}_i\right) + E\left(\hat{p}_i^2\right) c_{i+1}. \tag{16}$$

Proof. If X and Y are two independent random variables, it is simple to show that $Var\left(XY\right) = Var\left(X\right) E\left(Y^2\right) + Var\left(Y\right) E^2(X)$. In our model $\hat{p}_i$ and $\hat{m}_{i+1}$ are independent, and $E\left(\hat{m}_{i+1}\right) =$

$\mu_{i+1} - t_{i+1}$. Then $c_i = Var\left(\hat{m}_i\right) = Var\left(\left(\Delta t_{i+1} + \hat{m}_{i+1}\right)\hat{p}_i\right) = Var\left(\hat{m}_{i+1}\right)E\left(\hat{p}_i^2\right) + Var\left(\hat{p}_i\right)\left(\Delta t_{i+1} + E\left(\hat{m}_{i+1}\right)\right)^2 = (\mu_{i+1} - t_i)^2 Var\left(\hat{p}_i\right) + E\left(\hat{p}_i^2\right)c_{i+1}$. □

In analogy with (7), we have that $\int_{t_i}^{t_k} S(t)dt = (\mu_{i+1} - t_i)S(t_i)$, or that $\mu_{i+1} - t_i = \int_{t_i}^{t_k} S(t|t_i)dt$, and then $(\mu_{i+1} - t_i)^2\, Var\left(\hat{p}_i\right) = (\int_{t_i}^{t_k} S(t|t_i)\,dt)^2 \frac{p_i q_i}{N_i} = (\int_{t_i}^{t_k} S(t|t_{i-1})\,dt)^2 \frac{1}{p_i^2}\frac{p_i q_i}{N_i} = (\int_{t_i}^{t_k} S(t|t_{i-1})\,dt)^2 \alpha_i$, where $\alpha_i = \frac{q_i}{p_i N_i}$. Then

$$c_i = (\int_{t_i}^{t_k} S(t|t_{i-1})\,dt)^2 \alpha_i + p_i^2\left(1+\alpha_i\right)c_{i+1}. \tag{17}$$

The sample analog of (17) is

$$C_i = (t_{i+1}^* - t_i^*)^2 \frac{p_i}{Y_i q_i} + \hat{p}_i^2\left(1+\hat{\alpha}_i\right)C_{i+1},$$
$$i = k-1, k-2, \ldots, 0. \tag{18}$$

If we compare (18) with (14), we see that p_i^2 in (14) is replaced by $\hat{p}_i^2\left(1+\hat{\alpha}_i\right)$ in (18). We will soon see where this difference comes from.

The only left to do is to expand the recursive formula (17). Let us introduce the notation $A\left(t_i\right) = \int_{t_i}^{t_k} S(t)\,dt = S(t_{i-1})\int_{t_i}^{t_k} S(t|t_{i-1})dt$. When we apply recursion, we get $(S(t_{-1}) = S(0-) = 1))$

$$\begin{aligned} c_0 &= Var\left(\hat{\mu}_0\right) = A^2\left(t_0\right)\alpha_0 + p_0^2\left(1+\alpha_0\right)c_1 \\ &= A^2\left(t_0\right)\alpha_0 + A^2\left(t_1\right)\left(1+\alpha_0\right)\alpha_1 + S^2(t_1)\left(1+\alpha_0\right)\left(1+\alpha_1\right)c_2 \\ \ldots &= \sum_{i=0}^{k-1} A^2\left(t_i\right)\left[\prod_{j=0}^{i-1}\left(1+\alpha_j\right)\right]\alpha_i + S^2(t_{k-1})\prod_{j=0}^{k-1}\left(1+\alpha_j\right)c_k \\ &= \sum_{i=0}^{k-1} A^2\left(t_i\right)\left[\prod_{j=0}^{i-1}\left(1+\alpha_j\right)\right]\alpha_i, \end{aligned} \tag{19}$$

as $c_k = 0$ (or $A\left(t_k\right) = 0$). Let us introduce the "bias function"

$$\begin{aligned} B(t) = B\left(t_i\right) &= \prod_{j=0}^{i}\left(1+\alpha_j\right) - 1 \\ &= \prod_{0\le t_j \le t}\left(1+\alpha_j\right) - 1, t_i \le t < t_{i+1}. \end{aligned} \tag{20}$$

Then $\Delta B\left(t_i\right) = B\left(t_i\right) - B\left(t_{i-1}\right) = \prod_{j=0}^{i}\left(1+\alpha_j\right) - \prod_{j=0}^{i-1}\left(1+\alpha_j\right) = [\prod_{j=0}^{i-1}\left(1+\alpha_j\right)]\alpha_i = (B\left(t_{i-1}\right)+1)\alpha_i$, and

$$\begin{aligned} Var\left(\hat{\mu}\right) &= \sum_{i=0}^{k-1} A^2\left(t_i\right)\Delta B\left(t_i\right) = \sum_{i=0}^{k} A^2\left(t_i\right)\Delta B\left(t_i\right) \\ &= \int_{0-}^{t_k} A^2\left(t\right)dB\left(t\right). \end{aligned} \tag{21}$$

For Riemann-Stieltjes (R-S) integral, slightly simplified for our needs, see Appendix C. Notice that $A\left(t\right)$ is a continuous function, and (A 2) applies. If we use the approximation $B\left(t_i\right) \approx B_0\left(t_i\right) = \sum_{j=0}^{i}\alpha_j$, and $\Delta B_0\left(t_i\right) = \alpha_i$, we get the formula

$$Var\left(\hat{\mu}\right) \approx Var_0\left(\hat{\mu}\right) = \sum_{i=0}^{k-1} A^2\left(t_i\right)\alpha_i = \int_{0-}^{t_k} A^2\left(t\right)dB_0\left(t\right). \tag{22}$$

It is common to use the approximation in this estimation, so the K-M formula for estimating variance of the sample mean is

$$\begin{aligned} \hat{V}ar_0\left(\hat{\mu}\right) &= \int_{0-}^{t_k} \hat{A}^2\left(t\right)d\hat{B}_0\left(t\right) = \sum_{i=0}^{k} \hat{A}^2\left(t_i\right)\Delta\hat{B}_0\left(t_i\right) \\ &= \sum_{i=0}^{k-1}\left[\int_{t_i}^{t_k}\hat{S}\left(t\right)dt\right]^2\hat{\alpha}_i, \end{aligned} \tag{23}$$

where, again, for $i = k$, $\frac{0}{0} = 0$. A common method to derive the K-M estimator (9) is to start with the well-known *Greenwood's formula* for $Var[\hat{S}(t)]$ (see, e.g., O'Quigley 2008) but we don't need it here.

From $\Delta\hat{B}\left(t_i\right) = (1+\hat{B}\left(t_{i-1}\right))\Delta\hat{B}_0(t_i)$, it follows that

$$\begin{aligned} \hat{V}ar\left(\hat{\mu}\right) &= \hat{V}ar_0(\hat{\mu}) + \sum_{i=1}^{k}\hat{A}^2(t_i)\hat{B}(t_{i-1})\Delta\hat{B}_0(t_i) \\ &= \hat{V}ar_0(\hat{\mu}) + \int_0^{t_k}\hat{A}^2(t)\hat{B}(t)d\hat{B}_0(t). \end{aligned} \tag{24}$$

Obviously, from (24), $\hat{V}ar\left(\hat{\mu}\right) \ge \hat{V}ar_0(\hat{\mu})$ For the data in our examples, $\hat{V}ar\left(\hat{\mu}\right) = 73.6598$ is just slightly larger than $\hat{V}ar_0\left(\hat{\mu}\right) = 72.6588$. This should not be surprising after some analysis of $\hat{V}ar\left(\hat{\mu}\right) - \hat{V}ar_0(\hat{\mu})$ in (24). Our simulation in Section 8 shows comparable results, so, it seems the approximation $\hat{V}ar\left(\hat{\mu}\right) \approx \hat{V}ar_0(\hat{\mu})$ works nicely. We must note that the formula for $\hat{V}ar\left(\hat{\mu}\right)$ without approximation is practically never mentioned or used in applications, so we have it here as a "bonus." The whole point of this section was to introduce our estimators (especially for $Var(\hat{\mu})$) following a simple stochastic model, as a basis for initial understanding and more advanced study. A simple relationship between $\hat{\sigma}^2 = V_0$ and $\hat{V}ar(\hat{\mu})$ is described in Appendix B and is considered in simulation. It can be used to reduce the bias of $\hat{\sigma}^2$

Note 15. A variation of Greenwood's formula is proposed by Breuer (1982), where α_js are replaced by their *unbiased estimators* (under our model). Breuer also gives a formula for an unbiased estimator of the covariances $Cov\left(\hat{S}_i, \hat{S}_j\right)$ It seems the idea first appears in Irwin (1949). The Breuer's method can be used to construct an unbiased estimator for $Var(\hat{\mu})$.

8. Simulation Example

Our goal in this section is to illustrate how our estimators of μ, σ^2, and $Var(\hat{\mu})$ behave in a simulated example under different sample sizes and levels of censoring. We simulated samples from a Weibull distribution, with censoring, and compared the estimated values with their theoretical values. A problem is how to design censoring. We were thinking about two options

A. To use independent exponential censoring time, with variation of parameters.
B. To use a time window, where entry time is designed by a Poisson process and censoring time is the window end time. Or, for entry times we can use order statistics from uniform distribution. (progressive censoring)

Table 5. Estimating variance of the sample mean by simulation.

n	a	$\hat{\mu}$	$\hat{\sigma}^2$	$\hat{V}_0(\hat{\mu})$	$\hat{V}(\hat{\mu})$	$\hat{S}^2(\hat{\mu})$	$\hat{\sigma}^2/n$	$\widehat{\sigma}_0^2 = \tilde{V}_0$	R_c%	% cen.	% last*
25	20	17.6253	49.5231	3.7572	3.9537	4.6795	1.9809	53.2802	47.28	55.96	19.55
25	30	17.6787	52.2495	3.2247	3.3871	3.7712	2.0900	55.4743	35.19	42.84	13.34
25	40	17.7280	53.6848	2.9709	3.1118	3.2095	2.1474	56.6557	27.72	34.58	9.58
25	50	17.7328	53.9406	2.8034	2.9302	3.0057	2.1576	56.7440	23.04	29.06	7.57
25	60	17.7515	54.3588	2.7063	2.8252	2.9089	2.1744	57.0650	19.66	25.00	6.40
50	20	17.6852	53.4041	2.0685	2.1556	2.3902	1.0681	55.4726	48.36	55.98	18.31
50	30	17.7152	54.9275	1.7010	1.7623	1.7690	1.0985	56.6285	35.42	42.84	11.99
50	40	17.7181	55.6119	1.5454	1.5957	1.6164	1.1122	57.1572	28.03	34.69	8.67
50	50	17.7185	55.6469	1.4447	1.4881	1.5481	1.1129	57.0916	22.96	29.09	6.63
50	60	17.7470	56.0800	1.3946	1.4342	1.4566	1.1216	57.4746	19.58	25.04	5.83
100	20	17.7178	55.6670	1.0901	1.1239	1.1849	0.5567	56.7571	48.93	55.86	16.94
100	30	17.7389	56.4241	0.8782	0.8993	0.8995	0.5642	57.3023	35.75	42.95	10.61
100	40	17.7330	56.5906	0.7861	0.8021	0.8036	0.5659	57.3766	28.01	34.72	8.09
100	50	17.7409	56.6905	0.7363	0.7499	0.7558	0.5669	57.4269	23.01	29.11	6.54
100	60	17.7425	56.6106	0.7030	0.7150	0.7135	0.5661	57.3136	19.47	25.02	4.83
400	20	17.7387	57.1920	0.2824	0.2863	0.2862	0.1430	57.4744	49.38	55.90	13.85
400	30	17.7439	57.2892	0.2229	0.2249	0.2260	0.1432	57.5121	35.74	42.88	8.68
400	40	17.7443	57.3703	0.1991	0.2005	0.1991	0.1434	57.5695	27.97	34.71	6.21
400	50	17.7490	57.5208	0.1865	0.1877	0.1868	0.1438	57.7073	22.90	29.08	5.30
400	60	17.7431	57.4523	0.1781	0.1791	0.1789	0.1436	57.6304	19.35	24.97	4.21
1600	20	17.7458	57.5502	0.0712	0.0716	0.0722	0.0360	57.6214	49.51	55.90	12.42
1600	30	17.7459	57.5783	0.0561	0.0562	0.0563	0.0360	57.6343	35.82	42.93	8.10
1600	40	17.7433	57.5748	0.0499	0.0501	0.0505	0.0360	57.6247	27.94	34.70	5.91
1600	50	17.7492	57.6461	0.0467	0.0468	0.0468	0.0360	57.6929	22.88	29.08	4.60
1600	60	17.7438	57.5826	0.0446	0.0447	0.0440	0.0360	57.6272	19.34	24.99	3.99

(* % of samples with the last value censored)

In our example we applied simpler option A and used T with $W(2.5, 20)$ distribution. Option B is left to the reader. The theoretical mean and variance are $E(T) = 20 \times \Gamma(1 + 1/2.5) = 17.74528$, $\sigma = 7.593331$, $\sigma^2 = 57.65868$, so σ is relatively large. We simulated censoring times with exponential distribution with $\lambda = 1/a$, where $a = 20, 30, 40, 50, 60$, so the censoring is expected to decrease with larger a. $a = 10$ appeared to be too small, as some samples were completely censored when the number of repetitions was large. We used sample size $n = 25$, 50, 100, 400, and 1600 in one simulation run. Simulation runs are repeated 10,000 times for every combination of parameters, and the estimates $\hat{\mu}$, $\hat{\sigma}^2$, $\hat{V}_0(\hat{\mu})\ldots$ are averaged over them. It means that the reported values (e.g., for $\hat{\mu}$) are good Monte Carlo estimates for their true expected values (e.g., for $E(\hat{\mu})$) for the given combination of parameters. There are no ties ($d_0 = 0, d_i = 1, i > 0$) as we use a continuous distribution. The results are presented in Table 5 and illustrated in Figures 3 and 4.

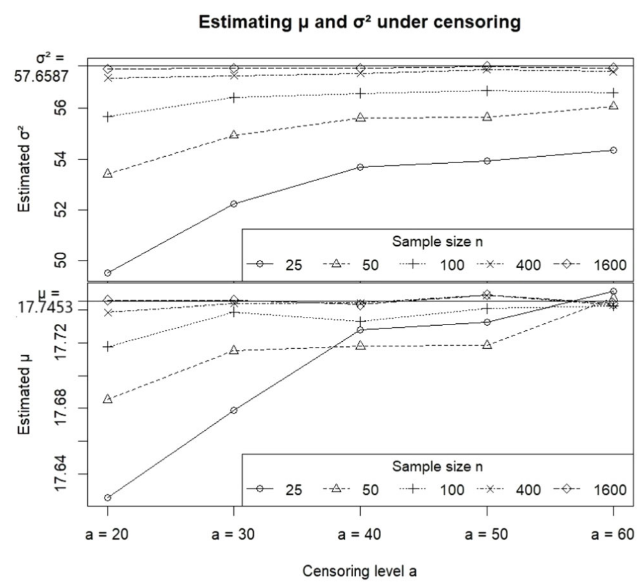

Figure 3. Estimating μ and σ^2 under different censoring levels a, and different sample sizes $n$using simulation.

Comments: The results are obtained under the model A of random (independent) censorship, and Weibull distribution of lifetimes, so the comments apply to this situation. $\hat{\mu}$ estimates μ accurately, on average (with minor bias for small n), and $\hat{\sigma}^2$ underestimates σ^2, less with larger samples. $\hat{V}_0(\hat{\mu}) = \hat{V}ar_0(\hat{\mu})$ underestimates true $Var(\hat{\mu})$, at least in smaller samples, if we do comparison with the empirical variance estimate $\hat{S}^2(\hat{\mu})$ of $Var(\hat{\mu})$ instead. $\hat{S}^2(\hat{\mu})$ is calculated as the sample variance of 10,000 repetitions of the appropriate $\hat{\mu}$, which is an unbiased estimator of $Var(\hat{\mu})$ (or, a Monte Carlo estimate). All other reported estimates are averaged over all 10,000 sample repetitions, as mentioned above. $\hat{V}\left(\hat{\mu}\right) = \hat{V}ar\left(\hat{\mu}\right)$ is calculated without using B_0 approximation (see Section 7), which somewhat decreases bias of $\hat{V}_0(\hat{\mu})$, but not significantly. In larger samples $\hat{V}\left(\hat{\mu}\right)$ is sometimes larger than $\hat{S}^2(\hat{\mu})$ and sometimes smaller but is close to it. An adjusted estimator $\widehat{\sigma}_0^2 = \tilde{V}_0 = \hat{\sigma}^2 + V_0(\hat{\mu})$ improves estimation of σ^2 as much as $\hat{V}_0(\hat{\mu})$ estimates $Var(\hat{\mu})$ properly (for details on $\widehat{\sigma}_0^2$ see Appendix B). $\hat{\sigma}^2/n$ shows what $\hat{V}ar(\hat{\mu})$ would be if there were no censoring, providing that $\hat{\sigma}^2$ is a good enough estimate of σ^2 ($\hat{\sigma}^2/n$ is "a plug-in comparison value," as the reviewer noticed); the proposed measure of the effect of censoring is given by R_c (see Note 14 and Example 4). R_c is consistent with the level of censoring defined by the parameter a, and also with the average % of the true censoring (real censoring of the last value counted in), but ought to be more sensitive to the position of censored observations. The average % of samples with the last value censored (and, hence, "corrected" to failure in estimation by Efron's method) is reported in the last

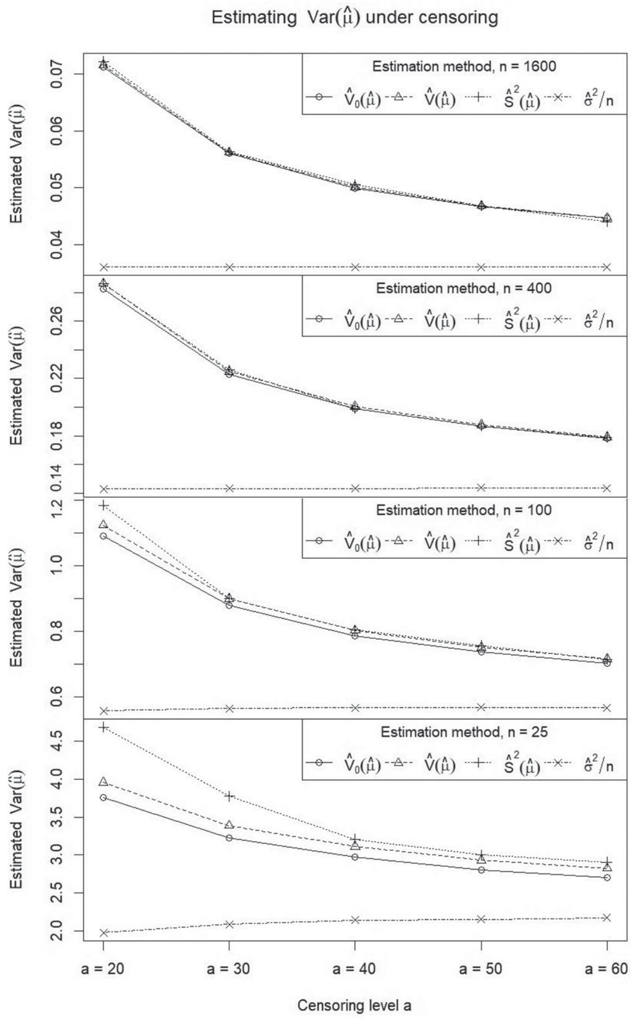

Figure 4. Estimating $Var(\hat{\mu})$ under different estimation methods, censoring levels a, and different sample sizes $n$using simulation.

column. For example, for the smallest sample size of $n = 25$ and the highest censoring level $a = 20$, the % of samples where the last value is censored is $\approx 20\%$. It seems this % has a small effect on estimation of μ, even in small sample sizes, and somewhat larger onσ^2 and $Var(\mu)$, but only in small sample sizes, which is consistent with the decreasing amount of censoring (increasing parameter a) and increasing sample size n, as shown in Figures 3 and 4. A more comprehensive simulation study would be welcomed, with other base distributions and censoring methods. Figure 3 illustrates the estimation of μ and σ^2 depending on the censoring levels and sample size and compares them with the theoretical values $\mu = 17.7453$ and $\sigma^2 = 57.6587$. The figure clearly shows that in this model, the estimates approach their target values of μ and σ^2, as the sample size increases and the censoring level decreases.

Figure 4 illustrates the estimation of $Var(\hat{\mu})$ depending on the method of estimation, censoring level and sample size, and compares it with their Monte Carlo calculated value $\hat{S}^2(\mu)$ on the top. A decreasing trend in the bias is evident in both figures, when censoring is reduced, or the sample size is increased. In Figure 4 we can see this if we start looking from the bottom portion of the graph to the top one, with increasing sample size, and checking the ratio of the estimated variance with $\hat{S}^2(\hat{\mu})$ as a reference value.

9. Discussion

When the sample $0 \le t'_1 \le t'_2 \le \cdots \le t'_n$ includes right-censored values, we apply a simple and intuitive method of adjustment—imputation to replace a censored value t'_i by y_i—the average of the values above t'_i, either uncensored or already adjusted, $y_i = \frac{1}{n-i}\sum_{j=i+1}^{n} y_j$. For the last, possibly censored value we use Efron's correction, $y_n = t'_n$, so that $\hat{\mu} = y_0 = \frac{1}{n}\sum_{j=1}^{n} y_j$. It is then obvious that only the uncensored values t'_i contribute directly to $\hat{\mu}$, with their weights a_i depending on the censoring information, that is, in $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} a_i t'_i$, $a_i = 0$ for a censored value, and $a_i > 0$ for an uncensored value. These weights are equal to the weights obtained by Efron's method of reweighting using an ad-hoc argument and can be used to introduce the K-M estimator of the survival curve as the probability mass function at the points of failure. If we consider a sequence of *different* uncensored times $0 = t_0 < t_1 < t_2 < \cdots < t_{k-1}$ and $t_k = t'_n$, (with appropriate counts of the ties and censored values), we see that the calculation of $\hat{\mu}$ can be reduced to a recursion $t^*_i = t_i\hat{q}_i + t^*_{i+1}\hat{p}_i$, $t^*_k = t_k$, $i = k-1, k-2, \ldots, 1, 0$, with $\hat{\mu} = t^*_0$. Essentially, t^*_i is the estimated total expected life for an element that survived up to t_i, or $t^*_i - t_i = \hat{E}(T - t_i | T \ge t_i)$. It appears that the sequence t^*_i is also useful in estimation of the population variance σ^2 and in all other cases related to estimation of $Var(\hat{\mu})$. More formal arguments related to the mean remaining life (MRL) can be established in using recursion for the sequence t^*_i. To explain the reasoning behind the "magic" formula for $\hat{V}ar(\hat{\mu})$ we used a simple stochastic model based on the main ideas from the K-M paper, combined with recursion. An interesting question remains—namely, why the "magic" formula derived here under a simple, restrictive model continues to apply in a much more general setting. Relative difference of the K-M estimator $\hat{V}ar_0(\hat{\mu})$ and $\hat{\sigma}^2/n$ can be used as a measure of the impact of censoring on the estimation of μ. A simple simulation study with a Weibull distribution is used to test how the estimators of μ, σ^2, and $Var(\hat{\mu})$ behave under different sample sizes and levels of censoring. A further simulation study with other distributions and different censoring methods should be used. It seems that further work on bias reduction in the estimation of $Var(\hat{\mu})$, particularly in small samples, is worthwhile. As a closing question, is it possible to extend this simple imputation-andrecursion framework to handle other forms of censoring? A partial answer is that the recursion *will work* when we can use a product limit type estimator for the survival function, such as with left-truncated and right-censored data.

Note 16. The reviewer suggested (see Section 2) to state explicitly what quantity is being estimated under the convention of Efron's correction to the right tail. In our case we want to estimate $\mu = E(T) = \int_0^\infty S(t)\,dt$ with an estimator $\hat{\mu}$, but we *are actually estimating* $E(\hat{\mu})$, which need not be equal to μ (the estimator can be biased). If we have an iid sample of T, without censoring, $E\left(\hat{\mu}\right) = E[\int_0^\infty \hat{S}(t)\,dt] = E\left(\bar{T}\right) = \mu$, but with censoring, $E\left(\hat{\mu}\right) = E[\int_0^\infty \hat{S}^E(t)\,dt] = \int_0^\infty E\left[\hat{S}^E(t)\right]dt =$

$\int_0^\infty E[\hat{S}^E(t) I(t < t_k)]dt$, where $\hat{S}^E(t)$ is the product limit estimator with Efron's correction, need not be equal to μ. Even under independent censorship, a formula for $E[S^E(t)]$ is quite complicated (Phadia and Shao 1999). At least, under a fairly general conditions (e.g., both lifetime and censoring variables with independent censoring are continuous, such as in our simulation), $E(\hat{\mu}) \leq \mu$, and $E(\hat{\mu}) \to \mu, n \to \infty$ (Gather and Pawlitschko 1998, Prop. 2.1). That is, $\hat{\mu}$ is assymptotically unbiased, with a downward bias; our simulation is consistent with this result.

Acknowledgments

The author would like to thank the anonymous reviewer for their comments, which greatly helped to improve the content and presentation of the paper.

Disclosure Statement

No potential conflict of interest was reported by the authors.

ORCID

Dragan Banjevic https://orcid.org/0000-0002-5549-0655

References

Aalen, O., Borgan, O., and Gjessing, H. (2008), *Survival and Event History Analysis: A Process Point of View*, New York, NY: Springer. [2]

Alvarez-Iglesias, A., Newell, J., Scarrott, C., and Hinde, J. (2015), "Summarising Censored Survival Data Using the Mean Residual Life Function," *Statistics in Medicine*, 34, 1965–1976. [2,3]

Banjevic, D. (2009), "Remaining Useful Life in Theory and Practice," *Metrika*, 69, 337–349. [3]

Borkowf, C. B. (2005), "A Simple Hybrid Variance Estimator for the Kaplan–Meier Survival Function," *Statistics in Medicine*, 24, 827–851. [2]

Bouaziz, O. (2023), "Fast Approximations of Pseudo-Observations in the Context of Right Censoring and Interval Censoring," *Biometrical Journal*, 65, 2200071. [2]

Breuer, J. (1982), "A Note on the Greenwood Variance Estimator in Cohort Life Tables: Isolating the Effect of One Source of Bias," *Communications in Statistics—Theory and Methods*, 11, 2947–2951. [10]

Brown, J., Hollander, M., and Kornwar, R. (1974), "Nonparametric Tests of Independence for Censored Data with Applications to Heart Transplant Studies," in *Reliability and Biometry, Statistical Analysis of Lifelength*, Philadelphia, PA: SIAM, pp. 327–354. [3]

Cole, S. R., Edwards, J. K., Naimi, A. I., and MuñOz, A. (2020), "Hidden Imputations and the Kaplan-Meier Estimator," *American Journal of Epidemiology*, 189, 1408–1411. [2]

Collett, D. (2003), *Modelling Survival Data in Medical Research* (2nd ed.), Boca Raton, FL: Chapman & Hall/CRC.

Datta, S. (2005), "Estimating the Mean Life Time Using Right Censored Data," *Statistical Methodology*, 2, 65–69. [1,2,4,6]

De Felice, F., Mazzoni, L., and Moriconi, F. (2023), "An Expectation-Maximization Algorithm for Including Oncological Covid-19 Deaths in Survival Analysis," *Current Oncology (Toronto, Ont.)*, 30, 2105–2126. [2]

De Felice, F., Mazzoni, L., Musio, D., Tombolini, V., and Moriconi, F. (2022), "Survival Analysis Using the Covid-Death Mean-Imputation (CoDMI) Algorithm: A First Clinical Application in Radiation Oncology," *In Vivo (Athens, Greece)*, 36, 2986–2992. [2]

Dinse, G. E. (1985), "An Alternative to Efron's Redistribution-of-Mass Construction of the Kaplan-Meier Estimator," *The American Statistician*, 39, 299–300. [4]

Efron, B. (1967), "The Two Sample Problem with Censored Data," in *Proceedings of the Fifth Berkeley Symposium in Mathematical Statistics* (vol. 4), New York, NY: Prentice-Hall, pp. 831–853. [2,3,5,6]

Efron, B. (1988), "Logistic Regression, Survival Analysis, and the Kaplan-Meier Curve," *Journal of the American Statistical Association*, 83, 414–425. [9]

Fisher, L., and Kanarek, P. (1974), "Presenting Censored Survival Data When Censoring and Survival Times May Not be Independent," in *Reliability and Biometry, Statistical Analysis of Lifelength*, Philadelphia, PA: SIAM, pp. 303–326. [5]

Fleming, T. R., and Harrington, D. P. (2005), *Counting Processes and Survival Analysis*, Hoboken, NJ: Wiley. [2]

Gather, U., and Pawlitschko, J. (1998), "On Efron's and Gill's Version of the Kaplan-Meier Integral," *Communications in Statistics—Theory and Methods*, 27, 181–192. [9,13]

Gillespie, B. W., Chen, Q., Reichert, H., Franzblau, A., Hedgeman, E., Lepkowski, J., Adriaens, P., Demond, A., Luksemburg, W., and Garabrant, D. H. (2010), "Estimating Population Distributions When Some Data Are Below a Limit of Detection by Using a Reverse Kaplan-Meier Estimator," *Epidemiology (Cambridge, Mass.)*, 21 Suppl 4, S64–S70. [2,4]

Gu, J., Fan, Y., and Yin, G. (2022), "Reconstructing the Kaplan-Meier Estimator as an M-Estimator," *The American Statistician*, 76, 37–43. [2]

Hosmer, D. W., and Lemeshow, S. (1999), *Applied Survival Analysis: regression Modeling of Time-to-Event Data* (2nd ed.), New York, NY: Wiley. [8]

Irwin, J. O. (1949), "The Standard Error of an Estimate of Expectation of Life," *Journal of Hygiene,* 47, 188–189. [9,10]

Kaplan, E. L., and Meier, P. (1958), "Nonparametric Estimation from Incomplete Observations," *Journal of the American Statistical Association*, 53, 457–481. [1,8]

Khan, H. N., Zaman, Q., Azmi, F., Shahzada, G., and Jakovljevic, M. (2022), "Methods for Improving the Variance Estimator of the Kaplan–Meier Survival Function, When There Is No, Moderate and Heavy Censoring-Applied in Oncological Datasets," *Frontiers of Public Health,* 10, 793648. [2]

Khan, M. H. R., and Shaw, J. E. H. (2016a), "Robust Bias Estimation for Kaplan-Meier Survival Estimator with Jackknifing," *Journal of Statistical Theory and Practice*, 10, 7–19. [3]

Khan, M. H. R., and Shaw, J. E. H. (2016b), "On Dealing with Censored Largest Observations under Weighted Least Squares," *Journal of Statistical Computation and Simulation*, 86, 3758–3776. [3]

Klein, J. P. L., and Moeschberger, M. L. (2003), *Survival Analysis: Techniques for Censored and Truncated Data* (2nd ed.), New York, NY: Springer. [2,7]

Kumazawa, Y. (1987), "A Note on an Estimator of Life Expectancy with Random Censorship," *Biometrika*, 74, 655–658. [8]

Kupka, J., and Loo, S. (1989), " The Hazard and Vitality Measures of Ageing," *Journal of Applied Probability*, 26, 532–542. [5]

Liang, W. (2022), "Random Weighting Estimation for Survival Function under Right Censorship," *Frontiers of Mathematics in China*, 17, 141–148. [2]

Malla, G., and Mukerjee, H. (2010), "A New Piecewise Exponential Estimator of a Survival Function," *Statistics & Probability Letters*, 80, 1911–1917. [2,3,5]

Mauro, D. (1985), "A Combinatoric Approach to the Kaplan-Meier Estimator," *The Annals of Statistics*, 13, 142–149. [5]

Meier, P. (1975), "Estimation of a Distribution Function from Incomplete Observations," *Journal of Applied Probability*, 12, 67–87. [5]

Mukangai, J. B., and Odongo, L. O. (2016), "Estimating Survivor Function Using Adjusted Product Limit Estimator in the Presence of Ties," *American Journal of Theoretical and Applied Statistics,* 5, 290–296. [2]

O'Quigley, J. (2008), *Proportional Hazards Regression*, New York, NY: Springer. [4,10]

Paukner, M., and Chappell, R. (2021), "Window Mean Survival Time," *Statistics in Medicine*, 40, 5521–5533. [2]

Phadia, E. G., and Shao, P. Y. (1999), "Exact Moments of the Product Limit Estimator," *Statistics & Probability Letters*, 41, 277–286. [13]

Qin, Y., Sasinowska, H., and Leemis, L. (2023), "The Probability Mass Function of the Kaplan–Meier Product–Limit Estimator," *The American Statistician*, 77, 102–110. [2]

Qin, G., and Zhao, Y. (2007), "Empirical Likelihood Inference for the Mean Residual Life under Random Censorship," *Statistics & Probability Letters*, 77, 549–557. [2,6]

Robins, J. M., and Rotnitzky, A. (1992), "Recovery of Information and Adjustment for Dependent Censoring Using Surrogate Markers," in *AIDS Epidemiology*, Boston, MA: Birkhäuser, pp. 297–331. [4]

Royston, P., and Parmar, M. K. B. (2013), "Restricted Mean Survival Time: An Alternative to the Hazard Ratio for the Design and Analysis of Randomized Trials with a Time-to-Event Outcome," *BMC Medical Research Methodology*, 13, 152. [2,6]

Ryan, T. P., and Woodall, W. P. (2005), "The Most-Cited Statistical Papers," *Journal of Applied Statistics*, 32, 461–474. [1]

Satten, G. A., and Datta, S. (2001), "The Kaplan–Meier Estimator as an Inverse-Probability-of-Censoring Weighted Average," *The American Statistician,* 55, 207–210. [2]

Shen, Y., Xie, M., and Tang, L. C. (2010), "Nonparametric Estimation of Decreasing Mean Residual Life with Type II Censored Data," *IEEE Transactions on Reliability*, 59, 38–44. [2]

Taylor, J. M. G., Murray, S., and Hsu, C. (2002), "Survival Estimation and Testing via Multiple Imputation," *Statistics & Probability Letters*, 58, 221–232. [2]

Wu, Y., and Kolassa, J. (2024), "Interval-Specific Censoring Set Adjusted Kaplan–Meier Estimator," *Journal of Applied Statistics*, 51, 2436–2456. [2]

Wu, Y., Kolassa, J., and Dong, N. (2025), "Restricted Mean Survival Time Based on Wu-Kolassa Estimator Compared to Kaplan-Meier Estimator," *Contemporary Clinical Trials*, 152, 107877. [2]

Zhao, H., Cheng, Y., and Bang, H. (2011), "Some Insight on Censored Cost Estimators," *Statistics in Medicine*, 30, 2381–2388. [2]